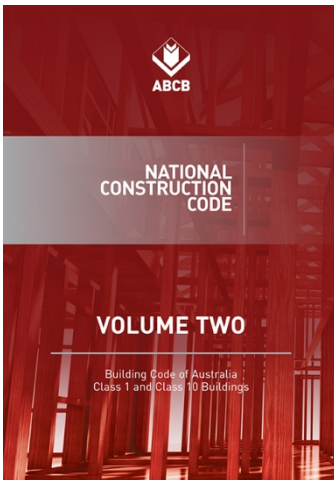


SINGLE STOREY HOUSE ENERGY EFFICIENCY

PBDS 2-002-C



The Performance Requirements of the National Construction Code (NCC) can be met using either a Performance Solution (previously known as an 'Alternative Solution') or a Deemed-to-Satisfy (DtS) Solution (complying with the DtS Provisions). The following demonstrates the performance based design process and should be used in conjunction with the Development of Performance Solutions Guidance document.

Scenario:

A single storey house is proposed to be built in Perth, Western Australia. Whilst the initial house design met the minimum glazing requirements for energy efficiency using the DtS Provisions, the client wishes to take advantage of cheaper, less energy efficient glazing that does not comply with the DtS Provisions. The client is also willing to improve other low cost energy efficiency elements to compensate for the glazing changes.



PREPARE PERFORMANCE-BASED DESIGN BRIEF

What are the design objectives?

- To reduce glazing costs for the client whilst maintaining the minimum level of energy efficiency performance required by the NCC.

Who should be consulted?

- Building designer, client, builder, energy assessor and the regulatory approval authority.

What is the basis of the Performance Solution?

- Using the Verification Method V2.6.2.2 in NCC Volume Two to provide flexibility in meeting the energy efficiency Performance Requirements.
- The acceptance criterion for the revised design is it using less heating and cooling energy than a 'reference building', which satisfies the minimum requirements of the relevant DtS provisions. This is based on V2.6.2.2.

What evidence is proposed?

- Output from energy rating software simulations which estimate the heating and cooling energy use for a 'reference building' and the revised design including the less energy efficient glazing and any other modifications needed to meet the minimum performance level through predicting less energy will be used.
- A written report explaining the approach used, the modifications made to the design and the heating and cooling energy use determined by the energy rating software simulations for both the 'reference building' and the revised design.

Which DtS Provisions are applicable?

- Parts 3.12.1, 3.12.2, 3.12.3 and 3.12.4 contain the DtS Provisions for the thermal performance of a dwelling relevant to energy efficiency.

Which Performance Requirement is applicable?

- The primary Performance Requirement is P2.6.1 – Building in Part 2.6 Energy Efficiency.

Note: for brevity, the applicable Performance Requirements have been limited. This solution may also impact other Performance Requirements and must be considered in accordance with 1.0.7.





CARRY OUT ANALYSIS, MODELLING OR TESTING

Which Assessment Methods are the most suitable and where can they be found?

Assessment Methods are listed in 1.0.5 and state that any Assessment Method or combination of them may be used to determine that a Performance Solution complies with the Performance Requirements. In this scenario, a Verification Method, V2.6.2.2 in BCA Volume Two, will be used as the Assessment Method.

The energy assessor has used the criteria detailed in V2.6.2.2 to establish the thermal performance of the 'reference building'. From these criteria, the 'reference building' has a level of performance equivalent to the minimum DtS provisions in Parts 3.12.1, 3.12.2, 3.12.3 and 3.12.4. The energy rating software that is being used meets the calculation method requirements identified in V2.6.2.2 and can provide separate heating and cooling energy use outputs. In modelling this scenario, the energy assessor also identified and recommended other low cost energy efficiency elements to compensate for the proposed glazing changes.



COLLATE AND EVALUATE RESULTS

The energy assessor developed a report detailing the modelling inputs and the energy rating done in accordance with the approach in V2.6.2.2. The report identified that the requirements of V2.6.2.2 had been addressed and included:

- Separate heating and cooling loads for the 'reference building' and the revised design
- Which elements were varied between the 'reference building' and the revised design
- Which energy rating software was used and how it met the calculation method requirements
- Which climate data was used
- Details demonstrating that the 'reference building' met the minimum requirements of the DtS Provisions in Parts 3.12.1, 3.12.2, 3.12.3 and 3.12.4.

The building designer, client and builder reviewed the report and agreed that the revised design satisfied the acceptance criteria and met the requirements of V2.6.2.2. This report was retained for submission to the regulatory approval authority as evidence of suitability that the Performance Requirement 2.6.1 had been satisfied.



PREPARE A FINAL REPORT

What should be in the final submission?

The final report provided by the energy assessor contained sufficient details to demonstrate that the requirements of V2.6.2.2, and subsequently the Performance Requirement P2.6.1 had been satisfied.

A covering letter was provided by the building designer to supplement this report summarising the key points relevant to this Performance Solution, which included:

- Scope of the solution, stakeholders involved, which Performance Requirement was assessed and the approach used
- Overview and outline of the modelling carried out by energy assessor
- Comparison of the heating and cooling use of the 'reference building' and the revised design with the acceptance criteria in V2.6.2.2
- Acceptance that the solution demonstrates compliance.

